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Outcome reporting in cohort studies of interventions

Guidance is needed to improve outcome definitions and reduce reporting bias

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Cohort studies of interventions complement randomised controlled trials. They are sometimes the best or the only evidence about interventions and outcomes that are difficult to assess in randomised controlled trials or populations that are underrepresented in these trials. Methodological advances have led to non-randomised designs that produce reliable causal estimates, especially cohort studies emulating target trials with new users and active comparators.¹

Reporting biases threaten the validity of randomised controlled trials and observational studies, and ongoing efforts are being made to improve registration and reporting of observational studies.^{2,3} Unfortunately, as Song and colleagues explain in their linked study ([doi:10.1136/bmj-2025-087975](https://doi.org/10.1136/bmj-2025-087975)), policies and procedures to minimise reporting biases in randomised controlled trials have not been applied widely to observational studies.⁴ For example, the International Committee of Medical Journal Editors (ICMJE) requires that all randomised controlled trials be registered,⁵⁻⁸ and primary outcomes must be defined in applications for funding and ethical approval. By contrast, the ICMJE does not require registration of observational studies, although some member journals encourage it. Observational studies might not involve sample size calculations that depend on well defined outcomes, and primary outcomes might not be prespecified in applications for funding or ethical approval.

To better understand the scale of the problem, Song and colleagues examine how outcomes were defined and reported in cohort studies of interventions registered on ClinicalTrials.gov.⁴ They found incomplete outcome definitions were common.⁴ Using a framework introduced by Zarin and colleagues,⁹ they explain there might be several ways to assess a concept or outcome “domain.” That is, the Zarin framework describes a complete outcome definition as having several elements, including specific measurement, specific metric, method of aggregation, and time point. Outcome definitions in randomised controlled trials and in observational studies that lack any of these elements might be compatible with many results,⁴ so registrations with incomplete outcome definitions might not prevent reporting biases. This framework is used by ClinicalTrials.gov and it has been incorporated in SPIRIT-2025 (Standard Protocol Items: Recommendations for Interventional Trials)^{10,11} and CONSORT-2025 (Consolidated Standards of Reporting Trials)^{12,13} reporting guidelines. However, it is not a perfect framework for meta-research because the elements that define an outcome are not always easy to define or assess. Moreover, little guidance exists to help investigators and meta-researchers determine

whether each element is defined completely for a given outcome. For example, it might be unclear what information is needed to define a “specific metric,” so researchers might disagree about whether the element is defined in an outcome. Moreover, this framework is not incorporated in guidelines for observational studies such as STROBE (Strengthening the Reporting of Observational Studies in Epidemiology)^{14,15}; published before Zarin et al) or TARGET (Transparent Reporting of Observational Studies Emulating a Target Trial)¹⁶ for target trial emulation studies. Assessing whether an outcome changed between a registration and a journal article reporting results is even more challenging. As in previous studies, Song and colleagues found that many outcomes differed between registrations and results reported in journal articles. It is a limitation that methods for identifying changes are not described in detail and appear to have relied on expert consensus rather than explicit criteria to determine what constituted a “change.” Practical guidance for defining outcomes, including what constitutes a change in an outcome, could support better registration and facilitate reproducible meta-research about the prevalence and consequences of outcome changes.

Reconciling contradictory guidance could also support better outcome registration. By following Zarin et al,⁹ Song and colleagues⁴ treat “method of aggregation” (ie, how data will be summarised within groups and compared between groups, using the mean or mean difference etc) as part of the outcome definition. Yet cohort study investigators might not consider “method of aggregation” necessary to define the outcome in each individual participant.¹⁷ Contemporary sources consider an outcome as one component of an “estimand,” the quantity a study aims to estimate.^{18,19} Other components of an “estimand” include population, treatment conditions, summary measure (to aggregate and compare outcomes), and handling intercurrent events.^{18,19} The estimand framework has been incorporated in guidance for observational studies from organisations such as the International Society for Pharmacoepidemiology and the International Society of Pharmacoeconomics and Outcomes Research,³ and in reporting guidelines such as TARGET.¹⁶ The Zarin framework predates the estimand framework, and it partly conflates outcome definitions (ie, the value for each individual participant or other unit of observation) with summary measure.

Song and colleagues found the prevalence of outcome switching (48%) in their sample of cohort studies fell within the wide range previously observed in randomised controlled trials (10-68%).⁴ However, the true prevalence in observational studies may be

even higher. The sample included only studies that were registered, which might be associated with better research practices compared with unregistered studies. Moreover, most included studies were prospective cohorts, which might be like randomised controlled trials insofar as investigators sometimes have control over data collection and therefore outcome definitions. In retrospective studies using existing data, researchers often have relatively less control over outcome definitions. Although Song and colleagues note that outcome switching was associated with the significance of results,⁴ consistent with the hypothesis that it is a source of reporting bias, outcomes are often changed for scientific or practical reasons without introducing bias. ClinicalTrials.gov allows investigators to update registrations many times, and makes all versions available publicly, which can help users understand the timing of outcome changes.

Song and colleagues found that investigators explained outcome changes in only two of 60 studies. Although CONSORT-2025^{12 13} recommends reporting outcome changes in randomised controlled trials, STROBE^{14 15} does not include standards for open science. The more recent TARGET guideline¹⁶ recommends reporting whether studies were registered, but it does not include a checklist item about outcome changes. Future updates and extensions of reporting guidelines for observational studies could mirror CONSORT recommendations for outcome changes.

In addition to outcome definitions, study protocols and statistical analysis plans should describe other components of estimands. Several registers allow researchers to upload these documents, but many registrations on ClinicalTrials.gov include only basic details that would be insufficient to verify or reproduce analyses. Improving the uptake of other open science practices could increase trust and reduce reporting biases in observational research.²⁰ Adopting open sciences practices could help address longstanding concerns about analytic flexibility and “P-hacking” in observational studies. Such concerns have led some journals to no longer publish studies based on certain cohorts. While some observational studies are purely descriptive or exploratory, broader adoption of open science practices might be especially important for cohort studies that test hypotheses. When doing research that aims to inform policy and influence patient care, we should expect investigators to register studies, and to share protocols and statistical analysis plans, before accessing or analysing data.

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